Plant disease classification

Fresh views and three paired seeds

Can crop and color augmentation improve field performance?

COMP 664 · Topic 2 · Fall 2026

Data and split integrity

Dataset	Purpose	Images
PlantVillage	Training	12,036
PlantVillage	Validation	1,364
PlantVillage	Clean test	3,351
PlantDoc	External test	742

Nine test classes; mosaic virus has no validation examples.
PlantDoc never enters optimization or checkpoint selection.

Six newly trained models

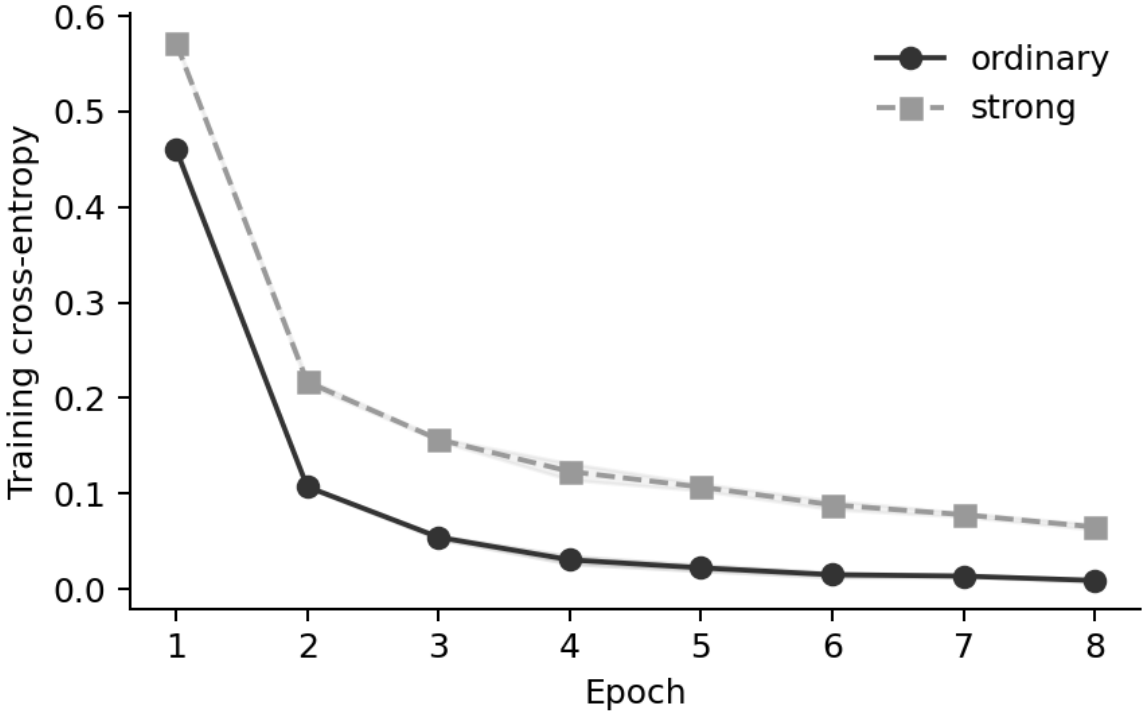
Three seeds × ordinary / strong augmentation × eight epochs.

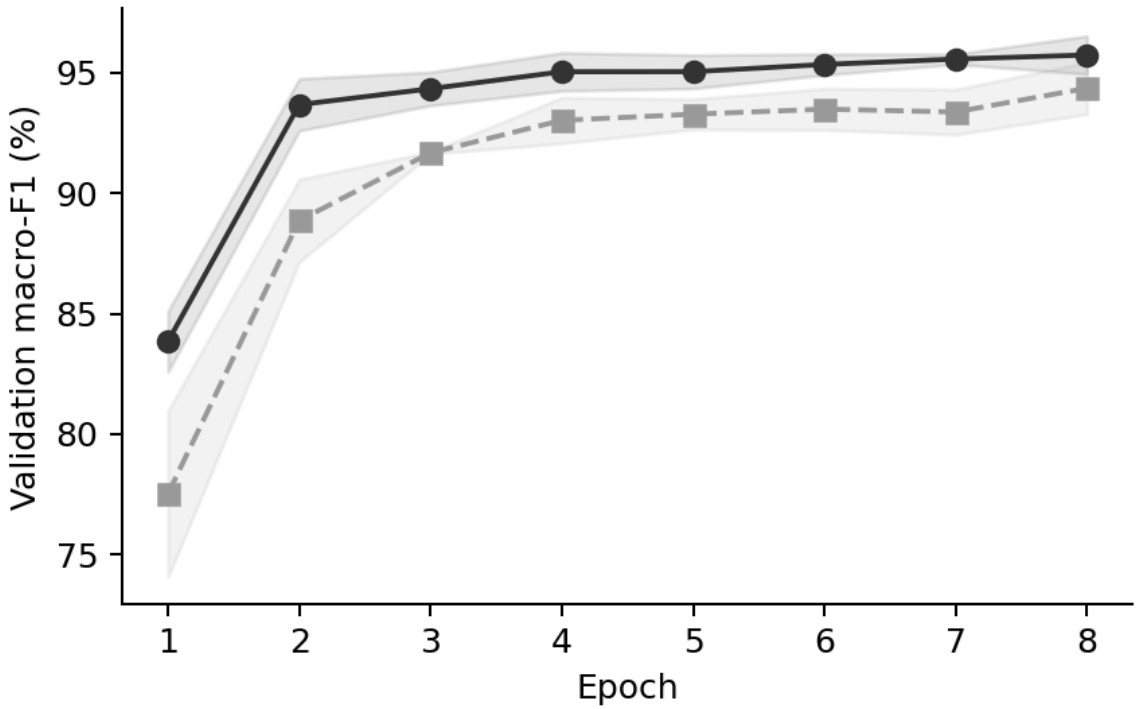
Fresh per-image transformation draws every epoch.
Only deterministic RGB resize is cached.

Same ImageNet MobileNetV3-Small, trainable blocks, AdamW,
batch size 96 and 1,008 steps per model.

All six checkpoints selected on source validation
before either test set is scored.

Training and validation history





48 completed epochs · 6,048 optimizer steps · 577,728 image presentations

Mean test results across three seeds

Metric (%)	Ordinary	Crop + color
Clean accuracy	97.26 ± 0.16	96.66 ± 0.58
Field accuracy	18.78 ± 1.10	20.13 ± 0.74
Clean macro-F1	96.32 ± 0.18	95.71 ± 0.50
Field macro-F1	7.76 ± 0.83	9.16 ± 0.73

Mean ± sample SD over seeds 664, 665 and 666.
Same 3,351 clean images and 742 field images for every run.

Every field result, with selected epochs

Seed	Training	Epoch	Accuracy (%)	Macro-F1 (%)
664	Ordinary	8	19.54	8.43
664	Crop + color	8	19.27	8.37
665	Ordinary	8	17.52	6.83
665	Crop + color	8	20.62	9.30
666	Ordinary	6	19.27	8.02
666	Crop + color	5	20.49	9.82

Selected epochs use PlantVillage validation only.

Paired field changes vary by seed

Seed	Accuracy change (pp)	Macro-F1 change (pp)
664	-0.27	-0.06
665	+3.10	+2.47
666	+1.21	+1.79
Mean ± SD	+1.35 ± 1.69	+1.40 ± 1.31

Two seed pairs improve; one declines slightly.
The mean gain is modest and not consistent across all seeds.

Mean per-class field recall

Class	Field n	Ordinary (%)	Strong (%)	Change (pp)
Healthy	63	0.5	0.5	+0.0
Early blight	88	34.8	26.5	-8.3
Bacterial spot	109	0.0	0.6	+0.6
Late blight	111	85.9	86.5	+0.6
Mosaic virus	54	0.0	0.0	+0.0
Yellow leaf curl	75	1.3	1.3	+0.0
Leaf mold	91	0.7	0.7	+0.0
Septoria	149	7.6	18.3	+10.7
Spider mites	2	0.0	0.0	+0.0

Field confusion matrices for seed 664



Seed 664 was fixed for illustration before field scoring.

Fixed demonstration photographs



Label: Early blight
Ordinary: Late blight
Strong: Late blight



Label: Early blight
Ordinary: Early blight
Strong: Late blight

Conclusion and remaining limitations

Mean field accuracy and macro-F1 improve modestly;
the benefit varies across the three seeds.

Remaining limits: missing validation class and leaf IDs,
near-duplicates, only two spider-mite field images.

A stronger next study needs independent field labels
and a predefined target-development/test separation.

Notebook and reproducibility evidence

Open Tomato_Project.ipynb for code, figures and saved outputs.

New method: experiment_fresh.py
Aggregate and audit: summarize_fresh.py

Six checkpoints, complete training histories, all test probabilities and the unchanged original experiment are included.